

Figure 1: Speech chain (Source: The Virtual Linguistic Campus (VLC) course on Speech Science)

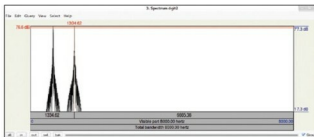


Figure 2: Spectrum for digit '2'

(DTMF) is the pair of frequencies assigned to each digit when we dial some number through the phone. Table 1 lists such frequencies corresponding to each digit. This is helpful in determining the dialled number. By Fourier analysis, we can obtain the frequencies present in a specific sound signal (dial tone). Figure 2 gives the frequency spectrum for the tone generated by pressing a specific digit.

In addition to DTMF, spectrum analysis is used to identify musical instruments, vowel identification and gender recognition, from audio signals.

Spectrogram analysis

A spectrogram represents three dimensions of a signal, i.e., time, frequency and amplitude. The horizontal and vertical dimensions indicate time (sec) and frequency (Hz) values, respectively. The darkness of the bands indicates the energy density or strength of the amplitude. Absence of darkness (pure white colour) represents the silence zone.

Spectrogram analysis is widely used in vowel identification, silence detection or formant analysis from specific speech utterances. Further, a spectrogram can also be used to identify the category or class of sounds (such as nasals, plosives, fricatives, etc.). A typical spectrogram is shown in Figure 3.

Formant analysis

A formant can be considered as a resonance of the vocal tract. In a typical spectrum envelope of a speech signal, formants depict picks of the envelope. Most of the vowels are characterised by the first two formants (F1 and F2) only.

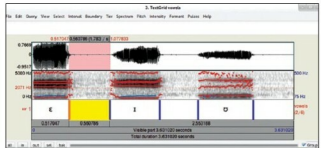


Figure 3: Spectrogram for three vowels

This is because formant frequencies reflect the articulatory gestures related to vowel height and the front-back position.

Format analysis is widely used in linguistics. There are many techniques to extract formants from a given speech signal. Let's discuss the easiest approach of getting Formant details from spectrograms. View the spectrogram of the signal and click on *Formant >> Show formants*, which will show the formant transitions in the spectrogram in red colour, as shown in Figure 3.

Praat

Praat is a tool designed for speech analysis. It was developed at the University of Amsterdam by Paul Boersma and David Weenink. According to them, Praat is a tool for “doing phonetics by computers.”

Downloading and installing Praat

There are multiple ways of getting Praat. Let us discuss the methods of downloading it on an Ubuntu 13.10 machine.

However, the process remains almost the same for all Linux distributions.

1. Standard way (through terminal):

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$ sudo apt-get install praat.
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2. Get it from the Ubuntu Software Center.
3. Get it from the Synaptic Package Manager.
4. Download it from the Praat website (Reference 2) and extract the archive.

Figure 4 shows a screenshot of Praat installed on the machine. Double clicking the Praat icon will open it. Alternatively, on a terminal, just typing *praat* will open its interface. The Praat icon comprises lips and ears, which exhibit the main speech production and perception organs in the human body.



Figure 4: Praat after installation